Unit 6 — Functions & *Linear Relationships*

ANSWER KEY

Answer key: every blank and answer is filled in green, with the steps.

🎯 By the end of this unit, I can…

- Decide whether a relation is a function
- Find and interpret slope and y-intercept in context
- Compare functions shown as graphs, tables, and equations
- Solve systems of linear equations

6.1 Functions

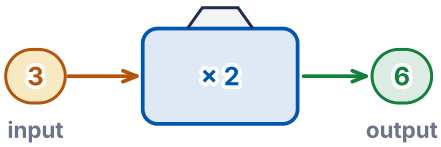
NC.8.F.1 — A FUNCTION IS A RULE THAT GIVES EACH INPUT EXACTLY ONE OUTPUT.

What is this section about? A function is like a vending machine. You press one button (the input). You get one snack (the output). If one button ever gave you two different snacks, the machine would be broken. That broken machine is *not* a function.

The Rule

A function gives each input exactly **one** output.

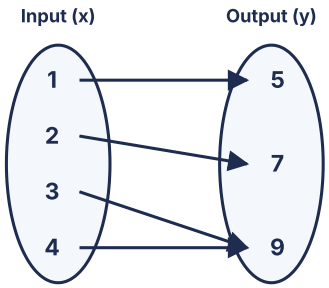
The input is usually **x**. The output is usually **y**.



Put 3 in. The rule "× 2" sends out 6.

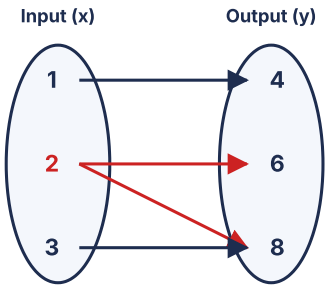
💡 **Outputs may repeat.** Two inputs can share the same output. Only an *input* with two different outputs breaks the rule.

See It: Mapping Diagrams



Function ✓

Each input has one arrow. 3 and 4 share the output 9. That is fine.

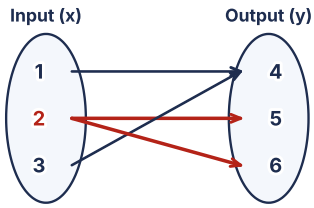


Not a function ×

The input 2 has two arrows: one to 6 and one to 8.

EXAMPLE

Worked example: Is this a function? (1, 4), (2, 5), (3, 4), (2, 6)

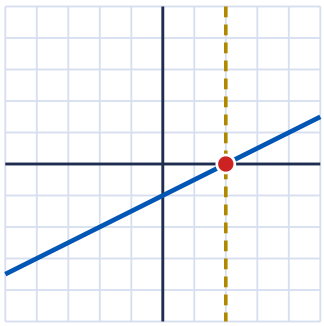


The input 2 has two arrows (to 5 and to 6).

Step 1. List the inputs (the first numbers): 1, 2, 3, 2.
Step 2. Look for an input that shows up twice. The input 2 shows up twice.
Step 3. Check its outputs. (2, 5) and (2, 6): the input 2 has two different outputs.
Step 4. So this is **not** a function.

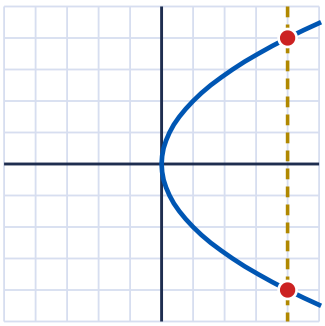
The Vertical Line Test (for graphs)

Slide a vertical line across the graph. If it ever touches the graph more than **once**, the graph is not a function.



Function ✓

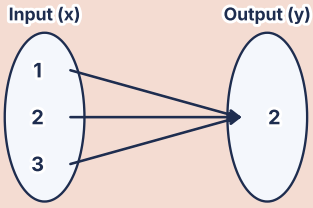
Every vertical line touches it once.



Not a function ×

The line at x = 4 touches it twice.

Find the mistake. Jay says (1, 2), (2, 2), (3, 2) is *not* a function, because the 2 shows up three times. Is Jay right? Write yes or no: **no**



Jay's points as a mapping diagram. Count the arrows leaving each input.

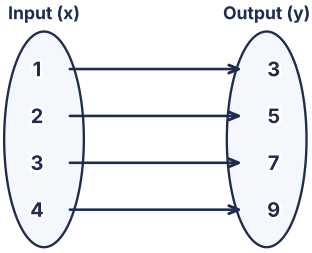
Why:

Jay is looking at the outputs. Outputs are allowed to repeat. Every input (1, 2 and 3) has only one output, so it IS a function.

TRY IT

Practice 1: Is this table a function? Write yes or no.

x	1	2	3	4
y	3	5	7	9



The table as a mapping diagram

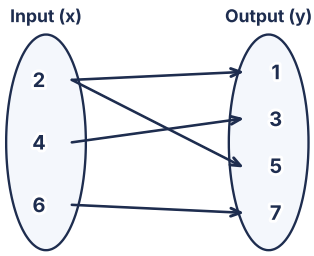
Function? yes

Why:

No input repeats, so each input has one output.

TRY IT

Practice 2: (2, 1), (4, 3), (2, 5), (6, 7). Function? Write yes or no.



Count the arrows leaving each input.

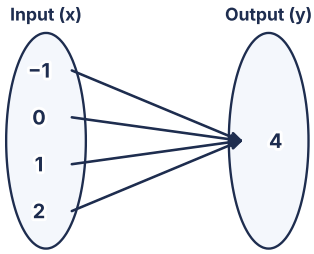
Function? no

Why:

The input 2 goes to 1 and also to 5. One input, two outputs.

TRY IT

Practice 3: (−1, 4), (0, 4), (1, 4), (2, 4). Function? Write yes or no.



Count the arrows leaving each input.

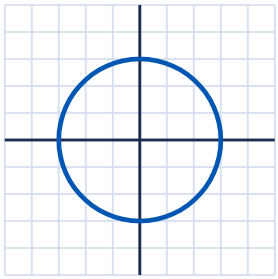
Function? yes

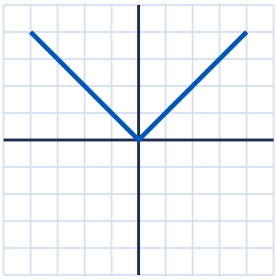
Why:

Every output is 4, but no input repeats. Repeated outputs are fine.

TRY IT

Practice 4 and 5: Use the vertical line test. Write yes or no for each graph.





Circle: function? **no**

V shape: function? **yes**

Why:

A vertical line at x = 0 touches the circle at the top and the bottom: two points.

Why:

Any vertical line touches the V only once.

✏ Practice: Functions

1. ★ Try it Is this a function? (3, 8), (5, 2), (3, 9), (6, 1)

○ Yes

● **No**

How to do it: The inputs are 3, 5, 3, 6. The input 3 shows up twice: (3, 8) and (3, 9). One input has two outputs, so it is **not** a function.

2. ★★ On level This table is **not** a function. Which input breaks the rule? Write just the number.

x	2	4	6	4
y	1	3	5	7

Answer: 4

How to do it: The inputs are 2, 4, 6, 4. The input 4 shows up twice. It goes to 3 and also to 7. So the input **4** breaks the rule.

3. ★★★ Challenge (1, 5), (2, 7), (☐, 9), (4, 11). Give one number for the box that makes this **NOT** a function.

Answer: 1, 2, or 4

How to do it: The inputs already used are 1, 2 and 4. If the box is one of them, that input gets a second output. Example: box = 1 gives (1, 5) and (1, 9). So 1, 2, or 4 works.

6.4 Linear Functions in Real-World Context

NC.8.F.4 — CONSTRUCT A FUNCTION TO MODEL A LINEAR RELATIONSHIP; INTERPRET RATE OF CHANGE AND INITIAL VALUE IN CONTEXT.

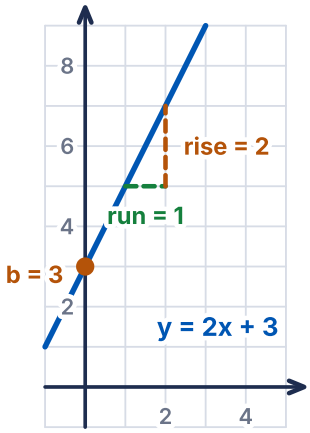
What is this section about? Real-world problems often follow a linear pattern — a steady rate of change plus a starting value. We can write equations in y = mx + b form to model these situations and use them to make predictions.

Slope & Y-Intercept in Context

In real-world problems, the slope and y-intercept have specific meanings:

m (slope) = the ___rate___ of change — how much y changes per unit of x.

b (y-intercept) = the ___starting___ value — what y equals when x = ___0___.



b is where the line crosses the y-axis. The triangle shows m: rise ÷ run = 2 ÷ 1 = 2.

💡 **Quick check:** The y-intercept is always the "before-anything-happens" value — the initial cost, starting amount, or fixed fee.

Keywords for Slope and Y-Intercept

Slope keywords (rate):

- per hour, per mile, per day
- each, every
- increases by, decreases by
- price/cost per item

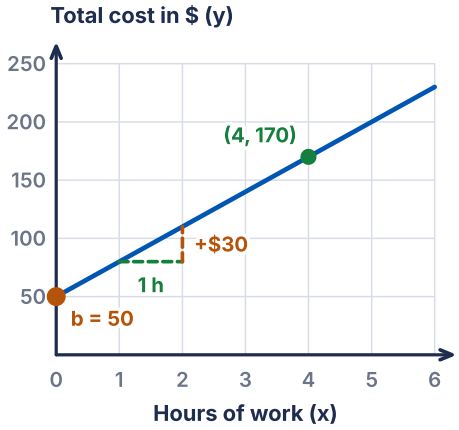
Y-intercept keywords (start):

- flat fee, base cost
- starting amount, initial
- up-front, deposit
- "to begin with"

EXAMPLE

Plumber example: A plumber charges $50 to come to your house, plus $30 per hour of work.

```
The flat fee ($50) is paid before any work happens → b = 50
The $30 per hour is the rate of change → m = 30
Equation: y = 30x + 50
```

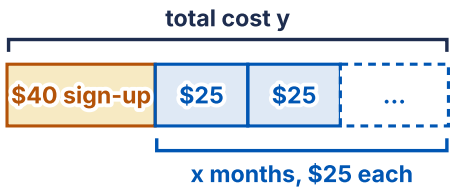


Start at $50 on the y-axis, then go up $30 for each hour.

Interpret: If x = 4 hours, then y = 30(4) + 50 = $170 total.

TRY IT

Practice 1: A gym charges a $40 sign-up fee plus $25 per month. Write the equation in slope-intercept form for the total cost *y* after *x* months.



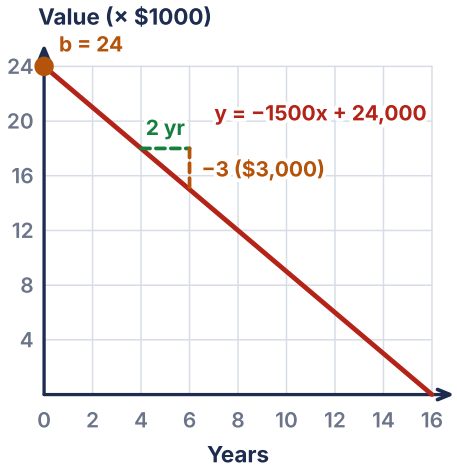
Which part is paid once? Which part repeats?

Slope (m) = 25 · Y-intercept (b) = 40

Equation: y = 25 x + 40

EXAMPLE

Car value example: The value of a car after x years is modeled by $y = -1500x + 24{,}000$.

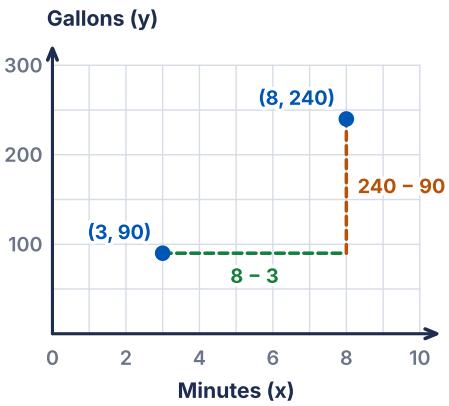


The line starts at b = 24 ($24,000 when new) and goes down $3,000 every 2 years ($1,500 per year).

`m = −1500 → the car loses` **`$1,500 per year`** `(slope is negative because value decreases)`
`b = 24,000 → the car was worth` **`$24,000`** `when new (when x = 0)`

TRY IT

Practice 2: A swimming pool is being filled. After 3 minutes there are 90 gallons; after 8 minutes there are 240 gallons. The pool was empty before filling started.



Rise ÷ run gives gallons per minute.

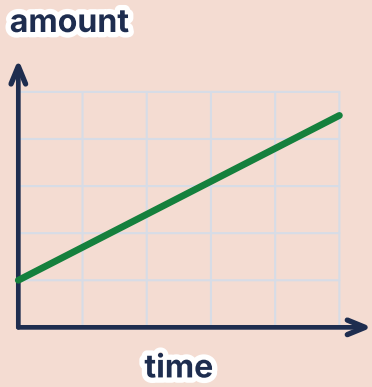
Find the slope (gallons per minute): m = 30

The y-intercept (initial gallons) = 0

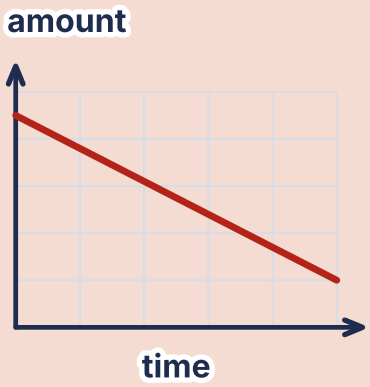
Equation: y = 30 x

Hint: slope = (240 − 90) ÷ (8 − 3) = 150 ÷ 5 = ?

⚠️ **Watch the sign of the slope:** If the quantity is *increasing* over time (filling, growing, earning), m is **positive**. If *decreasing* (depreciating, draining, using up), m is **negative**.



Filling: goes up, m > 0



Draining: goes down, m < 0

TRY IT

Practice 3: Your phone has 80% battery and loses 5% per hour of use. Write an equation for battery percent y after x hours.



Start at 80%. Lose 5% every hour.

m = −5 · b = 80

Equation: y = −5 x + 80

After 4 hours, battery = 60 %

✏ Practice: Linear Functions in Real-World Context

1. ★ Try it A taxi charges $3 to start, plus $2 per mile. What is the rate of change (slope)? Write just the number.

Answer: 2

How to do it: "Per mile" tells you the rate. Each mile adds $2, so m = 2. (The $3 to start is b.)

2. ★★ On level Bowling costs $4 for shoes, plus $5 per game. Write an equation for the total cost y after x games.

Answer: y = 5x + 4

How to do it: $5 per game is the rate → m = 5. The $4 for shoes is paid once → b = 4. So y = 5x + 4.

3. ★★★ Challenge A tutor charges a fixed fee plus a rate per hour. 2 hours cost $70. 5 hours cost $145. What is the fixed fee? Write just the number.

Answer: 20

How to do it: Rate: (145 − 70) ÷ (5 − 2) = 75 ÷ 3 = $25 per hour. 2 hours of work cost 2 × 25 = $50. Fixed fee: 70 − 50 = **$20**. Check: 25(5) + 20 = 145 ✓

6.5 Comparing Functions

NC.8.F.2 — COMPARE PROPERTIES OF TWO FUNCTIONS EACH REPRESENTED IN A DIFFERENT WAY (ALGEBRAICALLY, GRAPHICALLY, IN A TABLE, OR VERBALLY).

The big idea: Linear functions can be shown four ways — equation, table, graph, or verbal description. To compare two functions, get both into `y = mx + b` form (or just identify their slopes and y-intercepts). Then you can compare directly.

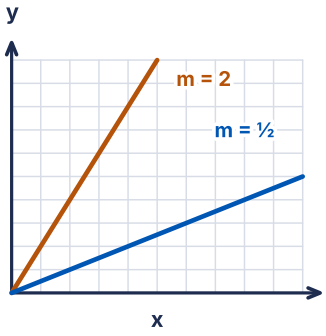
The Strategy

Before answering *any* comparison question, find the slope (m) and y-intercept (b) for **BOTH** functions first.

To compare *rate of change*, look at: **slope**

To compare *starting value*, look at: **y-intercept**

The function with the steeper graph has the **greater** slope.



Two lines from the same start. Compare how fast each one climbs.

EXAMPLE

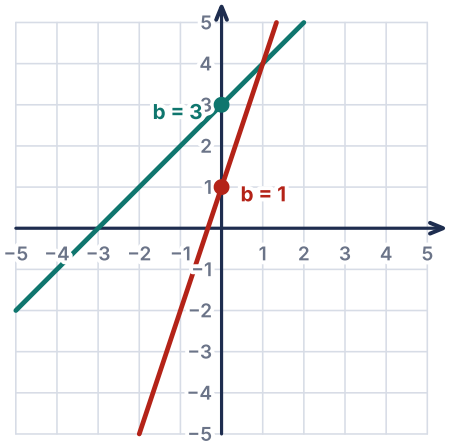
Compare these two functions:

Function A (equation):

y = x + 3

Function B (table):

x	0	1	2	3
y	1	4	7	10



A: y = x + 3 · B: y = 3x + 1 (steeper)

```
Function A: m = 1, b = 3
Function B: m = 3 (each row, y goes up by 3); b = 1 (when x = 0)
```

Greater rate of change? Function B (slope of 3 > slope of 1).

Higher starting value? Function A (b = 3 > b = 1).

TRY IT

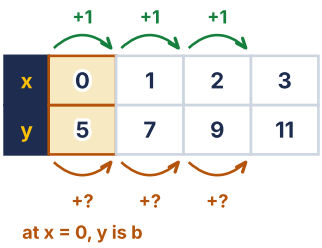
Practice 1: Compare these two functions:

Function A:

y = 4x + 2

Function B:

x	0	1	2	3
y	5	7	9	11



Function B: how much does y change each time x goes up 1?

Function A: m = 4 , b = 2

Function B: m = 2 , b = 5

Which has the higher rate of change? Function A

Which has the higher starting value? Function B

EXAMPLE

Cell phone plans: Plan A costs $20/month plus $0.05/minute. Plan B costs $30/month plus $0.03/minute.

```
Plan A: y = 0.05x + 20
Plan B: y = 0.03x + 30
At x = 0 minutes: Plan A = $20, Plan B = $30 → Plan A starts cheaper
At x = 500 minutes: Plan A = $45, Plan B = $45 → tied!
At x = 1000 minutes: Plan A = $70, Plan B = $60 → Plan B is cheaper for heavy use
```



Plan A starts lower but climbs faster than Plan B.

TRY IT

Practice 2: Two savings accounts:

- Account A starts with $300 and adds $25/week → y = 25x + 300
- Account B starts with $400 and adds $20/week → y = 20x + 400

Which has more money after 10 weeks?



Same 10 weeks, different start and different rate.

Account A after 10 weeks: y = 25(10) + 300 = $ 550

Account B after 10 weeks: y = 20(10) + 400 = $ 600

Account B has more after 10 weeks.

🔑 **Strategy:** When word problems compare two situations, write *both* equations first, then plug in the same value of x to compare. Or, if asked about long-term, look at the slopes — steeper slope wins eventually.

✏ Practice: Comparing Functions

1. ★ Try it Function A: y = 3x + 7. Function B: y = 5x + 1. Which has the greater rate of change?

- ○ Function A
- ● **Function B**

How to do it: Slopes: A has m = 3. B has m = 5. 5 > 3, so **Function B** has the greater rate of change. (The 7 and the 1 are starting values.)

2. ★★ On level Function A is the table. Function B is y = 4x + 9. Which has the greater rate of change?

x	0	1	2	3
y	4	10	16	22

- ● **Function A**
- ○ Function B
- ○ They are the same

How to do it: Table A: y goes up 6 each time x goes up 1, so m = 6. Function B: m = 4. 6 > 4, so **Function A** changes faster.

3. ★★★ Challenge Gym A costs $10 per visit. Gym B costs a $30 fee plus $5 per visit. After how many visits do they cost the same? Write just the number.

Answer: 6

How to do it: Gym A: y = 10x. Gym B: y = 5x + 30. Set them equal: 10x = 5x + 30 → 5x = 30 → x = **6**. Check: 10(6) = 60 and 5(6) + 30 = 60 ✓

6.6 Qualitative Graphs

NC.8.F.5 — DESCRIBE QUALITATIVELY THE FUNCTIONAL RELATIONSHIP BETWEEN TWO QUANTITIES BY ANALYZING A GRAPH; SKETCH A GRAPH FROM A VERBAL DESCRIPTION.

What's a qualitative graph? A graph that shows the *shape* of a relationship over time without exact numbers. The story of the graph matters more than specific values. Common examples: distance-time graphs, temperature-time graphs.

Reading the Shape

Each section of a qualitative graph tells you something. Going left-to-right, ask:

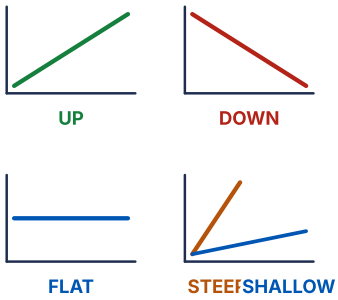
Going UP means the quantity is **increasing** .

Going DOWN means the quantity is **decreasing** .

FLAT (horizontal) means the quantity is **constant** (not changing).

STEEPER lines mean the change is **faster** .

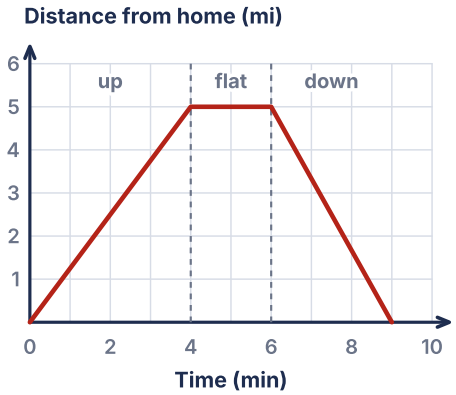
SHALLOWER (gentler) lines mean the change is **slower** .



The four shapes to look for in any graph.

EXAMPLE

Maria rides her bike to school:



Maria's Bike Ride

The graph shows Maria's distance from home over time. We can read the story:

- **0 to 4 min:** line goes UP at a steady rate → Maria is biking *away* from home toward school.
- **4 to 6 min:** line is FLAT → Maria stays at school (distance doesn't change).
- **6 to 9 min:** line goes DOWN steeper than it went up → Maria rides home *faster* than she went.

⚠️ **Common mistake:** When someone is *returning home*, the distance-from-home line goes DOWN (toward zero), not back up! The y-axis is "distance from start" — if you're getting closer to home, your distance is decreasing.

TRY IT

Practice 1: A swimmer dives into a pool, swims at a steady pace to the far wall, rests for a moment, then swims back *faster* than she went.

The graph below should show distance from start versus time. Match each phase to the correct description:



The swimmer's distance from the start

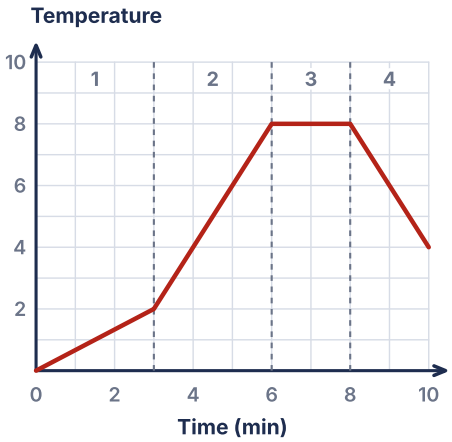
Phase 1 (going up at steady rate): she swam ___out___ to the far wall at a steady pace.

Phase 2 (flat line): she ___rested___ at the far wall.

Phase 3 (going down steeper): she swam ___back___ faster than she went.

EXAMPLE

A pot of water: A pot of water is placed on a stove. It heats slowly at first, then more rapidly, stays at boiling for several minutes, then is removed and cools gradually.



Phases 1 to 4 of the pot of water

- **Phase 1 (shallow up):** heats slowly
- **Phase 2 (steep up):** heats more rapidly
- **Phase 3 (flat):** at boiling, temperature stays constant
- **Phase 4 (down):** removed from stove, cools off gradually

TRY IT

Practice 2: Look at a distance-time graph with three phases:

- Phase 1 (0–3 min): steep upward line
- Phase 2 (3–6 min): flat line
- Phase 3 (6–10 min): gradual upward line



The three phases, drawn

Phase 1: the person is moving **fast** (away from start).

Phase 2: the person is **stopped** (not moving).

Phase 3: the person is moving **slowly** (still away from start).

📐 **Steps for sketching from a story:**

1. Break the story into 2–4 phases.
2. For each phase, decide: increase, decrease, or stay the same?
3. Decide if the change is fast (steep) or slow (shallow).
4. Connect phases at their endpoints — no jumps in the line!

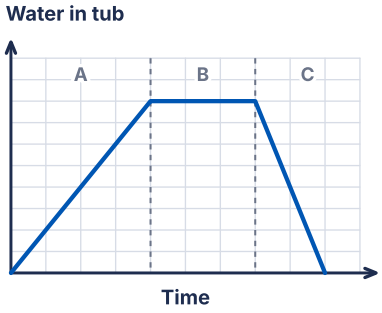
✏ Practice: Qualitative Graphs

1. ★ Try it A graph shows Leo's distance from home. The line goes DOWN. What is Leo doing?

- ○ Moving away from home
- ● **Moving toward home**
- ○ Standing still

How to do it: Going down means the distance from home is decreasing. Leo is getting closer to home, so he is **moving toward home**.

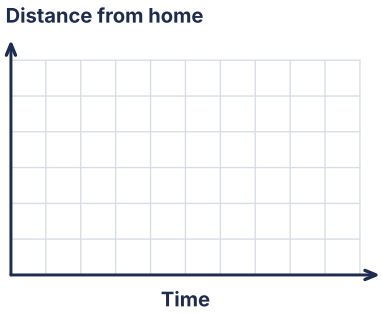
2. ★★ On level This graph shows the water in a bathtub. In which part is the tub draining?



- ○ Part A
- ○ Part B
- ● **Part C**

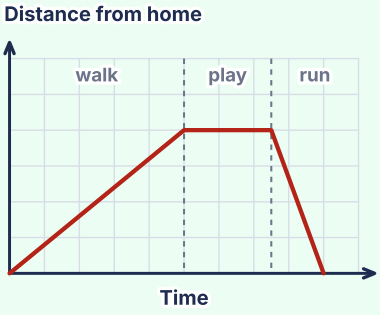
How to do it: Part A goes up (filling). Part B is flat (the water stays the same). Part C goes down, so the tub is draining in **Part C**.

3. ★★★ Challenge Sketch it: Sam walks slowly to the park. He plays there for a while. Then he runs home. Sketch his distance from home over time.



Answer: Gentle line up → flat → steep line down to 0

How to do it: Break the story into 3 parts. **Walks slowly away:** the line goes up, but not steep. **Plays at the park:** flat line. **Runs home:** the line goes down steeply, back to 0.



Mixed Practice: All Sections

NC.8.F.1 · NC.8.F.2 · NC.8.F.4 · NC.8.F.5 — REVIEW OF 6.1, 6.4, 6.5 AND 6.6

Put it all together. These questions mix every section. Start with the ★ ones if you want a warm-up. Stuck? Look back at the section's examples. You've got this!

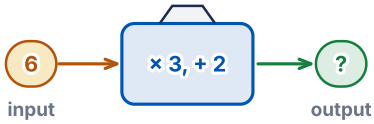
✏ Comprehensive Practice All sections

1. ★ Try it (6.1) Which set of points is a function?

- ○ (1, 2), (1, 3), (2, 4)
- ● **(0, 5), (1, 5), (2, 5)**
- ○ (4, 1), (4, 2), (4, 3)

How to do it: In the first set, 1 goes to 2 and to 3. In the last set, 4 goes to 1, 2 and 3. In (0, 5), (1, 5), (2, 5) every input is different, so it **is** a function. Outputs may repeat.

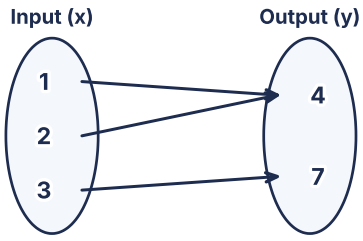
2. ★ Try it (6.1) A function machine has the rule "× 3, then + 2." What comes out when 6 goes in?



Answer: 20

How to do it: 6 × 3 = 18. Then 18 + 2 = **20**.

3. ★★ On level (6.1) Is this mapping a function?



- ● **Yes**
- ○ No

How to do it: Each input (1, 2, 3) has exactly one arrow. Two inputs share the output 4, and that is allowed. So **yes**, it is a function.

4. ★ Try it (6.4) A phone plan costs y = 15x + 35, where x is the number of months. What does the 35 mean?

- ○ The cost for each month
- ● **A one-time starting fee**
- ○ The number of months

How to do it: 35 is b, the y-intercept. It is the cost when x = 0 months, so it is **a one-time starting fee**. The 15 is the cost for each month.

5. ★★ On level (6.4) A candle is 30 cm tall. It burns down 2 cm each hour. Write an equation for its height y after x hours.

Answer: y = −2x + 30

How to do it: Start: 30 cm → b = 30. It loses 2 cm each hour → m = −2. So **y = −2x + 30**.

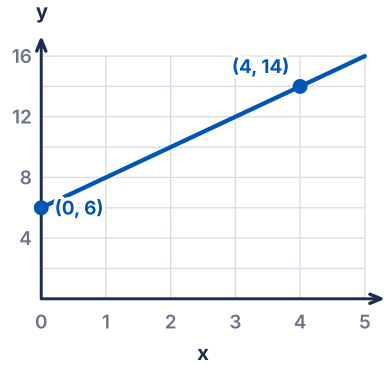
6. ★★★ Challenge (6.4) The table shows the money left on a gift card after x movies. After how many movies will the card have $0? Write just the number.

Movies (x)	0	1	2	3
Money left (y)	$50	$45	$40	$35

Answer: 10

How to do it: Each movie uses $5, so m = −5. The card starts at $50, so b = 50: y = −5x + 50. The card is at $0 when 5x = 50, so x = 50 ÷ 5 = **10** movies.

7. ★★ On level (6.5) Function A is the graph. Function B is y = 3x + 1. Which has the greater rate of change?



- ○ Function A
- ● **Function B**

How to do it: Graph A: from (0, 6) to (4, 14) the rise is 8 and the run is 4, so m = 8 ÷ 4 = 2. Function B: m = 3. 3 > 2, so **Function B** has the greater rate of change. (A has the greater start: 6 > 1.)

8. ★★★ Challenge (6.5) Store A rents bikes for $8 per hour. Store B charges a $12 fee plus $5 per hour. For a 6-hour ride, which store is cheaper?

- ○ Store A, by $6
- ● **Store B, by $6**
- ○ Store B, by $4
- ○ They cost the same

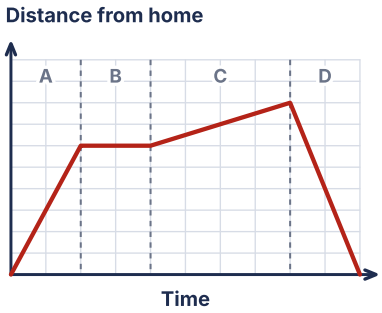
How to do it: Store A: y = 8(6) = $48. Store B: y = 5(6) + 12 = 30 + 12 = $42. 48 − 42 = 6, so **Store B is cheaper by $6**.

9. ★ Try it (6.6) On a graph of temperature over time, a flat part means the temperature is...

- ○ going up
- ○ going down
- ● **staying the same**

How to do it: Flat (horizontal) means no change. The temperature is **staying the same** (constant).

10. ★★ On level (6.6) The graph shows Ali's distance from home. In which part is Ali moving, but at his slowest speed?



- ○ Part A
- ○ Part B
- ● **Part C**
- ○ Part D

How to do it: Part B is flat, so Ali is stopped there. He moves in Parts A, C and D. Part C is the least steep (shallowest), so he moves slowest in **Part C**.

11. ★★★ Challenge (6.6) Find the mistake: Ben walks home from school. He draws his distance from home as a line going UP. What did Ben do wrong?

Answer: **His line should go DOWN, not up.**

How to do it: As Ben walks home, he gets closer to home. His distance from home gets smaller, so the line should go **DOWN** toward 0, not up.